

Section 2: Electricity

- a) Units
- b) Mains electricity
- c) Energy and potential difference in circuits
- d) Electric charge

a) Units

Students will be assessed on their ability to:

- 2.1 use the following units: ampere (A), coulomb (C), joule (J), ohm (Ω), second (s), volt (V), watt (W).

b) Mains electricity

Students will be assessed on their ability to:

- 2.2 understand and identify the hazards of electricity including frayed cables, long cables, damaged plugs, water around sockets, and pushing metal objects into sockets
- 2.3 understand the uses of insulation, double insulation, earthing, fuses and circuit breakers in a range of domestic appliances
- 2.4 understand that a current in a resistor results in the electrical transfer of energy and an increase in temperature, and how this can be used in a variety of domestic contexts
- 2.5 know and use the relationship:
power = current $\times$ voltage
$$P = I \times V$$

and apply the relationship to the selection of appropriate fuses
- 2.6 use the relationship between energy transferred, current, voltage and time:
energy transferred = current $\times$ voltage $\times$ time
$$E = I \times V \times t$$
- 2.7 understand the difference between mains electricity being alternating current (a.c.) and direct current (d.c.) being supplied by a cell or battery.

c) Energy and potential difference in circuits

Students will be assessed on their ability to:

- 2.8 explain why a series or parallel circuit is more appropriate for particular applications, including domestic lighting
- 2.9 understand that the current in a series circuit depends on the applied voltage and the number and nature of other components
- 2.10 describe how current varies with voltage in wires, resistors, metal filament lamps and diodes, and how this can be investigated experimentally
- 2.11 describe the qualitative effect of changing resistance on the current in a circuit
- 2.12 describe the qualitative variation of resistance of LDRs with illumination and of thermistors with temperature
- 2.13 know that lamps and LEDs can be used to indicate the presence of a current in a circuit
- 2.14 know and use the relationship between voltage, current and resistance:
voltage = current $\times$ resistance
 $V = I \times R$
- 2.15 understand that current is the rate of flow of charge
- 2.16 know and use the relationship between charge, current and time:
charge = current $\times$ time
 $Q = I \times t$
- 2.17 know that electric current in solid metallic conductors is a flow of negatively charged electrons
- 2.18 understand that:**
- **voltage is the energy transferred per unit charge passed**
 - **the volt is a joule per coulomb.**

d) Electric charge

Students will be assessed on their ability to:

- 2.19 identify common materials which are electrical conductors or insulators, including metals and plastics
- 2.20 describe experiments to investigate how insulating materials can be charged by friction**
- 2.21 explain that positive and negative electrostatic charges are produced on materials by the loss and gain of electrons**
- 2.22 understand that there are forces of attraction between unlike charges and forces of repulsion between like charges**
- 2.23 explain electrostatic phenomena in terms of the movement of electrons**
- 2.24 explain the potential dangers of electrostatic charges, eg when fuelling aircraft and tankers**
- 2.25 explain some uses of electrostatic charges, eg in photocopiers and inkjet printers.**

Section 3: Waves

- a) Units
- b) Properties of waves
- c) The electromagnetic spectrum
- d) Light and sound

a) Units

Students will be assessed on their ability to:

- 3.1 use the following units: degree (°), hertz (Hz), metre (m), metre/second (m/s), second (s).

b) Properties of waves

Students will be assessed on their ability to:

- 3.2 understand the difference between longitudinal and transverse waves and describe experiments to show longitudinal and transverse waves in, for example, ropes, springs and water
- 3.3 define amplitude, frequency, wavelength and period of a wave
- 3.4 understand that waves transfer energy and information without transferring matter
- 3.5 know and use the relationship between the speed, frequency and wavelength of a wave:

wave speed = frequency × wavelength

$$v = f \times \lambda$$

- 3.6 use the relationship between frequency and time period:

$$\text{frequency} = \frac{1}{\text{time period}}$$

$$f = \frac{1}{T}$$

- 3.7 use the above relationships in different contexts including sound waves and electromagnetic waves
- 3.8 understand that waves can be diffracted when they pass an edge**
- 3.9 understand that waves can be diffracted through gaps, and that the extent of diffraction depends on the wavelength and the physical dimension of the gap.**

c) The electromagnetic spectrum

Students will be assessed on their ability to:

- 3.10 understand that light is part of a continuous electromagnetic spectrum which includes radio, microwave, infrared, visible, ultraviolet, x-ray and gamma ray radiations and that all these waves travel at the same speed in free space
- 3.11 identify the order of the electromagnetic spectrum in terms of decreasing wavelength and increasing frequency, including the colours of the visible spectrum
- 3.12 explain some of the uses of electromagnetic radiations, including:
 - radio waves: broadcasting and communications
 - microwaves: cooking and satellite transmissions
 - infrared: heaters and night vision equipment
 - visible light: optical fibres and photography
 - ultraviolet: fluorescent lamps
 - x-rays: observing the internal structure of objects and materials and medical applications
 - gamma rays: sterilising food and medical equipment
- 3.13 understand the detrimental effects of excessive exposure of the human body to electromagnetic waves, including:
 - microwaves: internal heating of body tissue
 - infrared: skin burns
 - ultraviolet: damage to surface cells and blindness
 - gamma rays: cancer, mutation

and describe simple protective measures against the risks.

d) Light and sound

Students will be assessed on their ability to:

- 3.14 understand that light waves are transverse waves which can be reflected, refracted **and diffracted**
- 3.15 use the law of reflection (the angle of incidence equals the angle of reflection)
- 3.16 construct ray diagrams to illustrate the formation of a virtual image in a plane mirror
- 3.17 describe experiments to investigate the refraction of light, using rectangular blocks, semicircular blocks and triangular prisms

- 3.18 know and use the relationship between refractive index, angle of incidence and angle of refraction:

$$n = \frac{\sin i}{\sin r}$$

- 3.19 describe an experiment to determine the refractive index of glass, using a glass block
- 3.20 describe the role of total internal reflection in transmitting information along optical fibres and in prisms
- 3.21 explain the meaning of critical angle c
- 3.22 know and use the relationship between critical angle and refractive index:

$$\sin c = \frac{1}{n}$$

- 3.23 understand the difference between analogue and digital signals**

- 3.24 describe the advantages of using digital signals rather than analogue signals**

- 3.25 describe how digital signals can carry more information**

- 3.26 understand that sound waves are longitudinal waves and how they can be reflected, refracted **and diffracted**
- 3.27 understand that the frequency range for human hearing is
20 Hz – 20,000 Hz
- 3.28 describe an experiment to measure the speed of sound in air

- 3.29 understand how an oscilloscope and microphone can be used to display a sound wave**

- 3.30 describe an experiment using an oscilloscope to determine the frequency of a sound wave**

- 3.31 relate the pitch of a sound to the frequency of vibration of the source**

- 3.32 relate the loudness of a sound to the amplitude of vibration.**